

EE5311: Digital IC Design

Assignment 5 — NAND Gate Layout

Ishan Aji Hassan
Roll No: EE23B097

September 7, 2026

1 Part (a): NAND Gate Delay — Schematic-Level Simulation

1.1 Single NAND Cell: Layout Verification

The single NAND gate cell was laid out in Magic/KLayout and verified for design rule and layout-vs-schematic correctness before extraction.

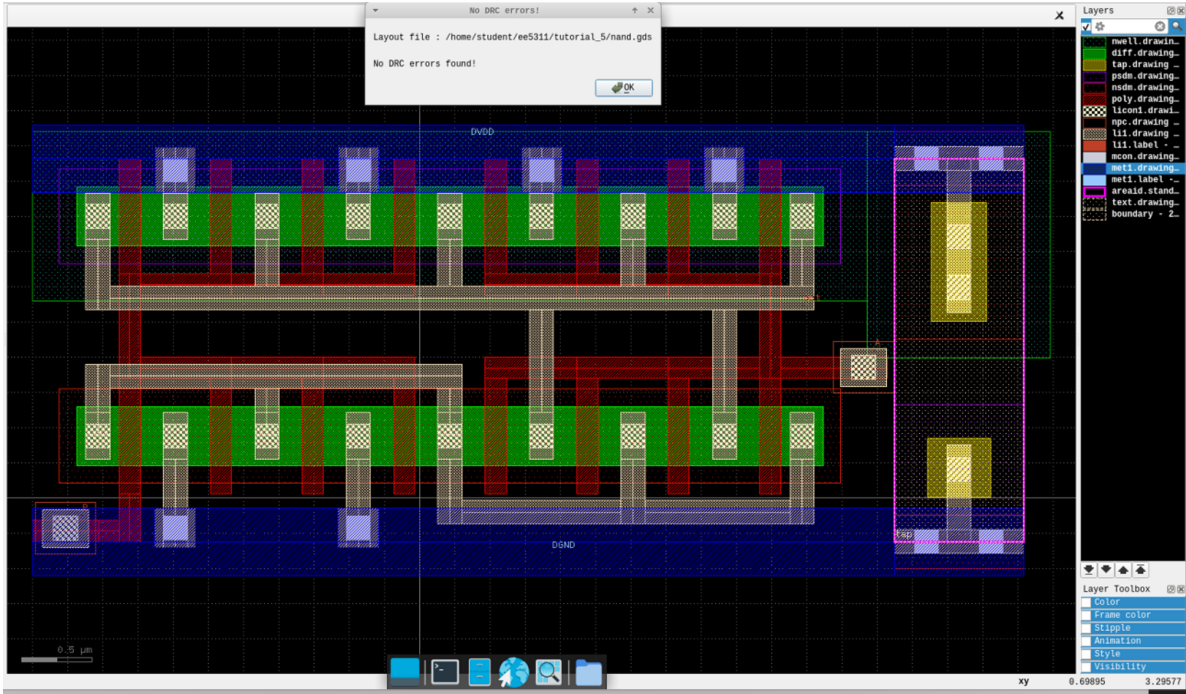


Figure 1: DRC report for the single NAND gate layout — no DRC errors found.

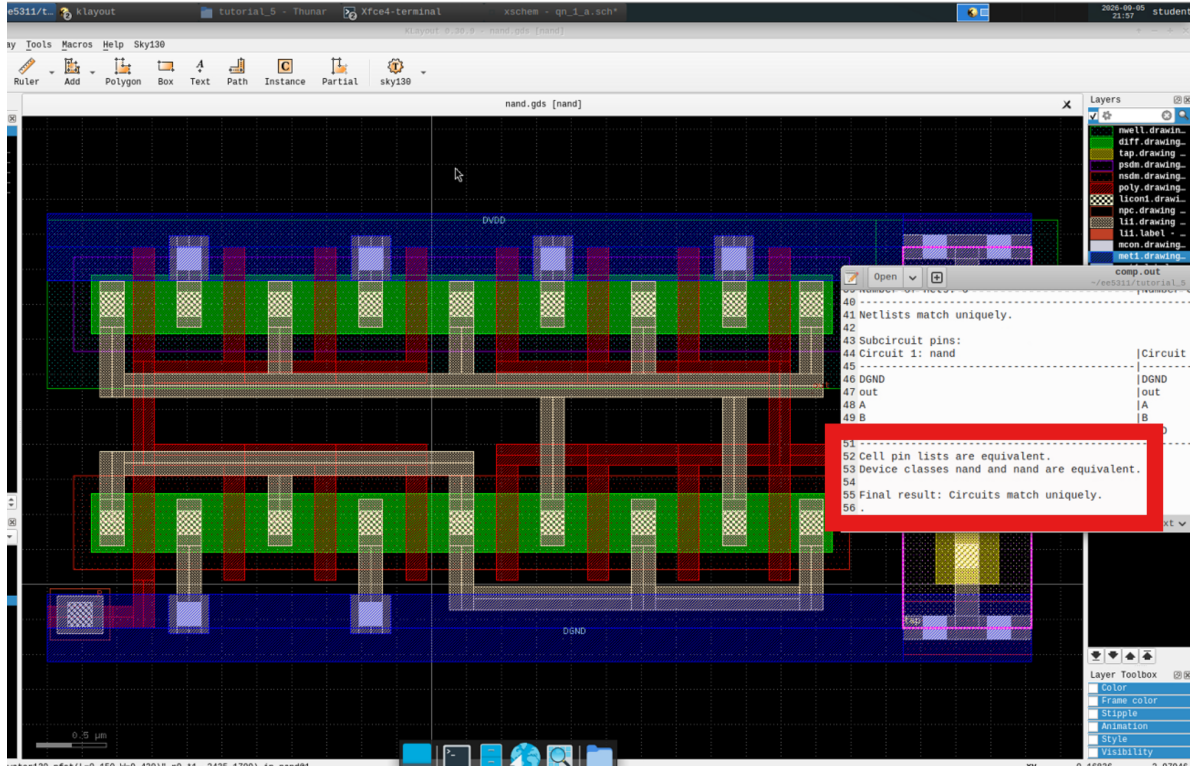


Figure 2: LVS report for the single NAND gate layout — netlists match uniquely against the schematic.

1.2 Delay Simulation

The extracted netlist of the single NAND cell (`nand_extracted.spice`) was instantiated twice in the xschem schematic (first NAND driving a second, identical NAND), and a transient simulation was run in ngspice with $V_{DD} = 1.8\text{ V}$.

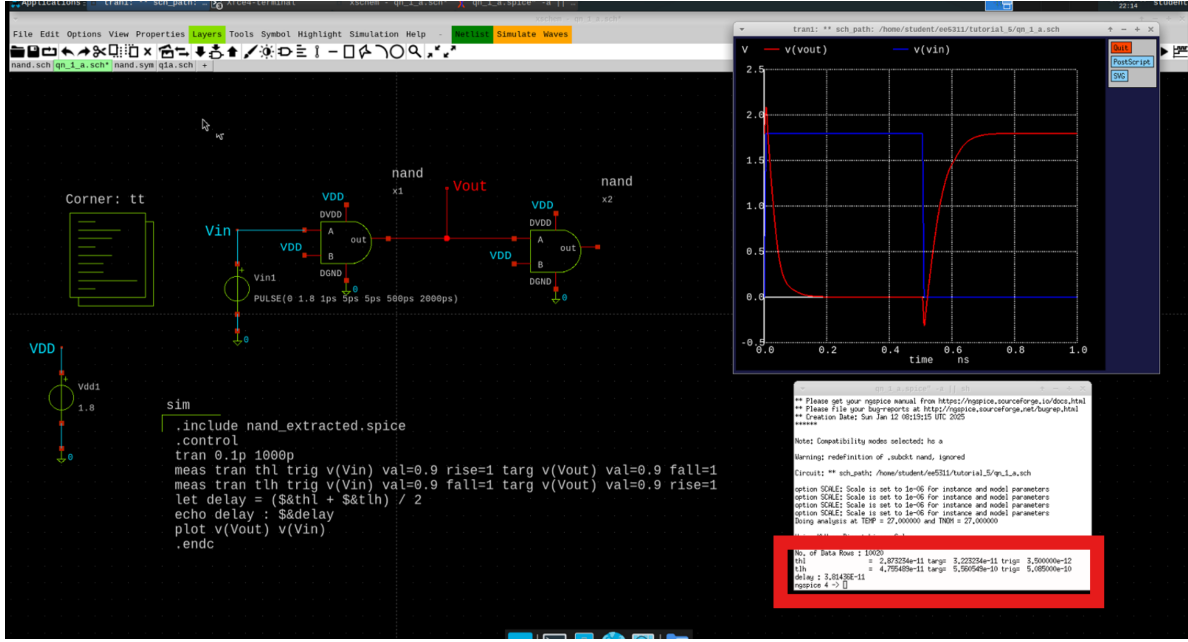


Figure 3: Schematic (two extracted-NAND instances), transient waveform, and delay measurement.

The measured propagation delays are:

Quantity	Value
t_{HL}	28.73 ps
t_{LH}	47.55 ps
Average propagation delay	38.14 ps

1.3 Comparison with Logical Effort

Delay = $(gh + p)t_p$. For the 2-input NAND gate ($W_n = W_p = 1.68 \mu\text{m}$, $2\times$ drive):

$$g = \frac{C_{\text{in, gate}}}{C_{\text{in, inv}(2\times)}} = \frac{1.68 + 1.68}{0.84 + 1.68} = \frac{4}{3}$$

$$h = \frac{C_{\text{load}}}{C_{\text{in, gate}}} = 1 \quad (\text{Driver and load are identical})$$

$$p = \frac{C_{\text{parasitic, gate}}}{C_{\text{parasitic, inv}(2\times)}} \approx \frac{\sum W_{\text{drain, gate}}}{\sum W_{\text{drain, inv}}} = \frac{1.68 + 1.68 + 1.68}{0.84 + 1.68} = 2$$

Input gate capacitance of unit inverter (C_{inv}):

$$\begin{aligned} C_{\text{inv}} &= (C_{\text{ox}}L + C_{\text{gso}})(W_n + W_p) \\ &= (8.34 \times 0.15 + 0.245) \times (0.42 + 0.84) \approx 1.89 \text{ fF} \end{aligned}$$

Using PDK values, the drain saturation currents for the unit inverter transistors are:

$$I_{DSAT,n} = \frac{1}{2} \mu_n C_{oxn} \left(\frac{W_n}{L_n} \right) (V_{DD} - V_{Tn})^2 \left(\frac{E_{Cn} L_n}{V_{DD} - V_{Tn} + E_{Cn} L_n} \right) \approx 164.60 \mu\text{A}$$

$$I_{DSAT,p} = \frac{1}{2} \mu_p C_{oxp} \left(\frac{W_p}{L_p} \right) (V_{DD} - |V_{Tp}|)^2 \left(\frac{E_{Cp} L_p}{V_{DD} - |V_{Tp}| + E_{Cp} L_p} \right) \approx 118.48 \mu\text{A}$$

Equivalent resistances:

$$R_{eq,n} \approx \frac{5}{7} \frac{V_{DD}}{I_{DSAT,n}} = \frac{5}{7} \left(\frac{1.8}{164.60 \times 10^{-6}} \right) \approx 7.81 \text{ k}\Omega$$

$$R_{eq,p} \approx \frac{5}{7} \frac{V_{DD}}{I_{DSAT,p}} = \frac{5}{7} \left(\frac{1.8}{118.48 \times 10^{-6}} \right) \approx 10.85 \text{ k}\Omega$$

$$R_{eq,avg} = \frac{R_{eq,n} + R_{eq,p}}{2} \approx 9.33 \text{ k}\Omega$$

Ideal unit inverter propagation delay time constant (t_p):

$$t_p = \ln(2) \cdot R_{eq,avg} \cdot C_{inv}$$

$$= \ln(2) \times 9.33 \text{ k}\Omega \times 1.89 \times 10^{-15} \text{ F} \approx \mathbf{12.22 \text{ ps}}$$

Total NAND gate delay:

$$\text{Delay} = (g \cdot h + p) \cdot t_p$$

$$= \left(\frac{4}{3} \times 1 + 2 \right) \times 12.22 \text{ ps} = \frac{10}{3} \times 12.22 \text{ ps} \approx \mathbf{40.73 \text{ ps}}$$

From simulation, delay = **38.14 ps** ($\sim 6.8\%$ error). The slight discrepancy is due to the finite input edge (5 ps) and the internal NMOS stack node being pre-discharged since input B is held at V_{DD} .

2 Part (b): Layout-Extracted Delay of the Two-NAND-Gate Layout

2.1 Two-NAND Layout (nand2): Layout Verification

A single combined layout containing both NAND gates, connected as required (output of the first driving input A of the second, with input B of both tied to V_{DD}), was drawn and verified.

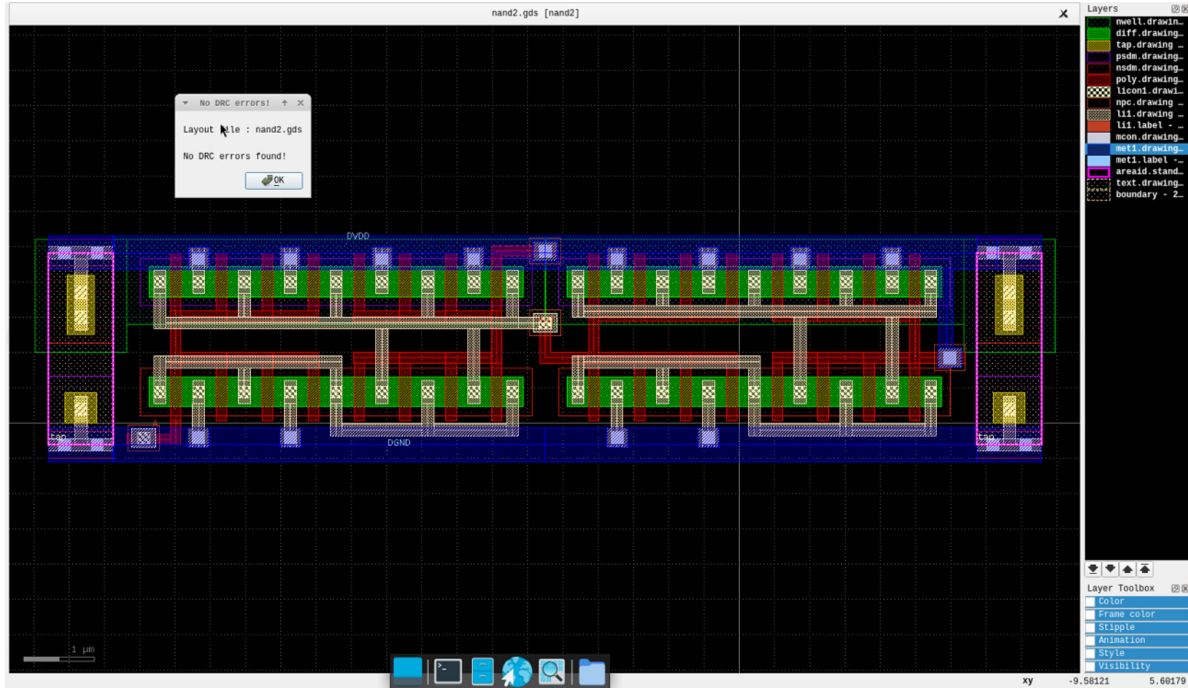


Figure 4: DRC report for the two-NAND-gate (nand2) layout — no DRC errors found.

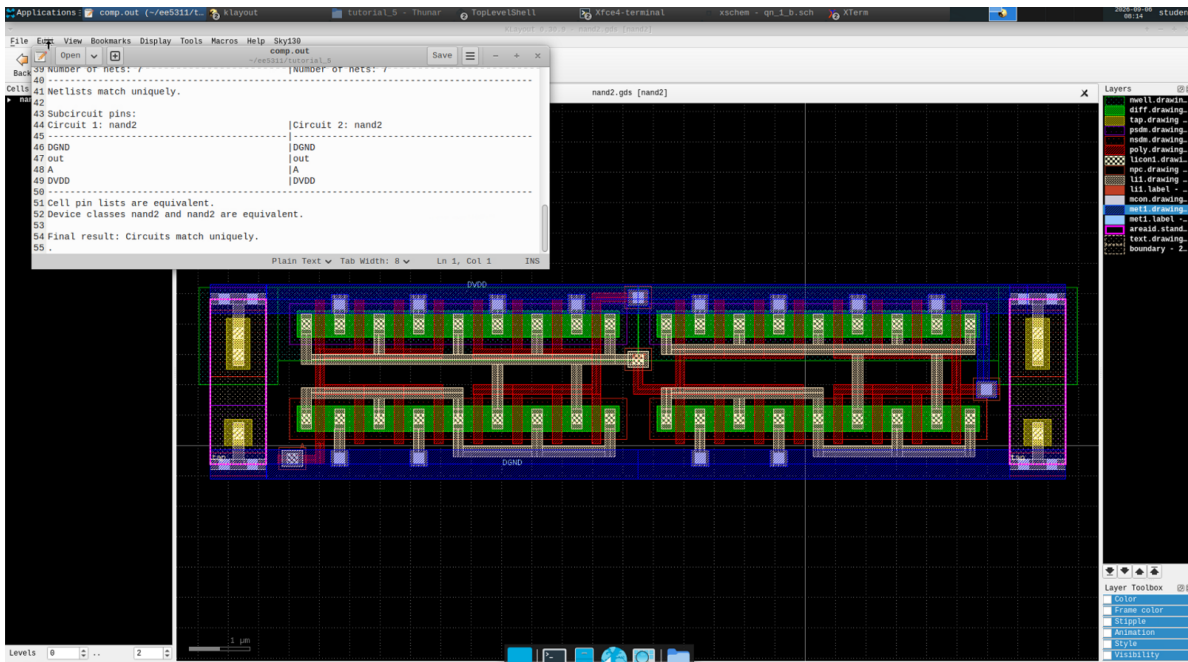


Figure 5: LVS report for the nand2 layout — netlists match uniquely against the schematic.

2.2 Delay Simulation

The fully parasitic-extracted netlist of the combined layout (nand2_extracted.spice) was instantiated once in the schematic and simulated under identical stimulus conditions.

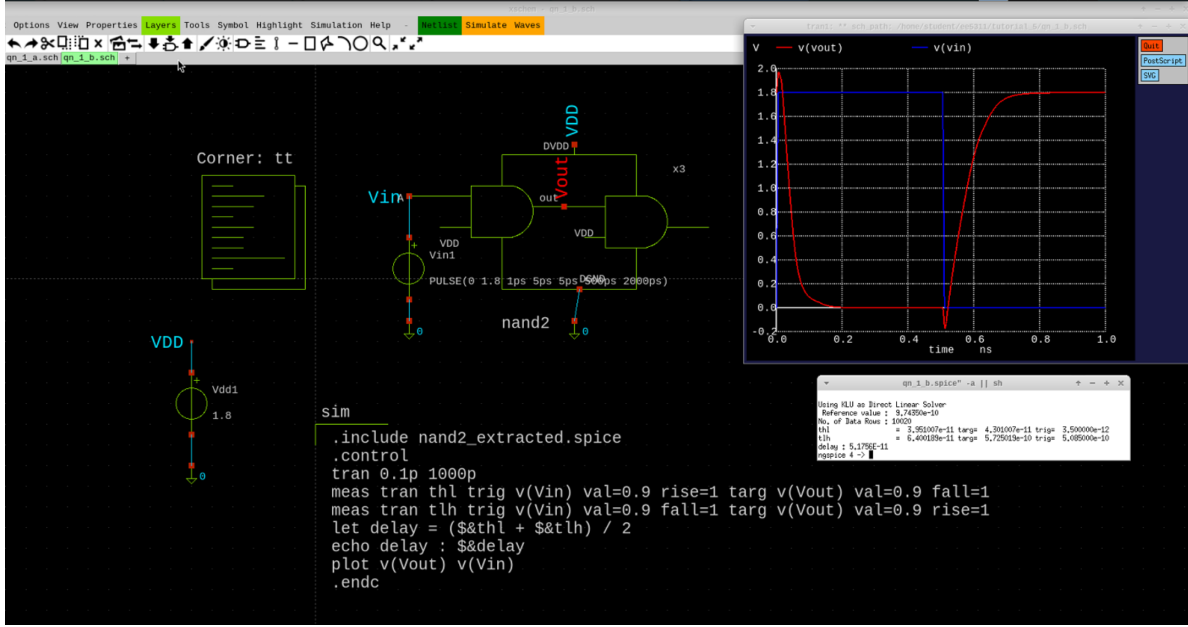


Figure 6: Schematic (single extracted `nand2` instance), transient waveform, and delay measurement.

The measured propagation delays are:

Quantity	Value
t_{HL}	39.51 ps
t_{LH}	64.00 ps
Average propagation delay	51.76 ps

3 Summary

Method	Average delay
Schematic (extracted single-NAND cell, 2 instances)	38.14 ps
Layout-extracted (combined 2-NAND layout)	51.76 ps
Logical effort estimate (assumed $\tau \approx 15$ ps)	≈ 50 ps

The layout-extracted delay (51.76 ps) is higher than the schematic-level estimate (38.14 ps) due to the additional interconnect parasitics (routing capacitance/resistance) captured only when both gates are extracted from a single, fully-routed layout. This value also aligns more closely with the logical-effort-based estimate.